

Project Synopsis

Project Title	Hyper-Local Cloud Library
TA Mentor	George Rahul
Document	DASS Project Concept Document
Creation Date	24th January, 2026
Version History	Final Version
Created and Worked on by	Guntesh Singh, Khushi Dhingra, Mohana Krishna, Nikhil M.C.B,Aryan Bharania
Client	Rahul, TheFounderPapa

Description:

The proposed solution entails creating a cloud-based, mobile-first library management ecosystem tailored specifically for young children aged 2 to 10. Designed to serve micro-libraries, communities, and schools, the platform is built utilizing the MERN stack and React Native Expo, functioning seamlessly across different mobile hardware from a single codebase. The system relies on a Parent-Child hierarchical account structure paired with a bifurcated UI. This design ensures an engaging, highly visual "Kids Mode" for children to independently explore books, while securely restricting cart management, checkout flows, and account settings to a PIN-gated "Parent Mode."

A central feature of the user experience is a Retrieval-Augmented Generation (RAG) recommendation system. Powered by a dedicated Node.js microservice utilizing cloud embedding models for vector search, this architecture acts as a conversational "virtual librarian." By employing rolling-window memory and strict contextual grounding, the AI interprets a child's requests and reading history to offer highly personalized, hallucination-free book recommendations without exceeding system token limits.

To support multi-branch operations efficiently, the backend utilizes a single master database with metadata filtering rather than fragmented databases. By integrating Expo location services, the platform queries the exact inventory of the user's nearest branch, guaranteeing that children only receive recommendations for physical books currently in stock at their local library. Furthermore, the e-commerce flow enforces a strict single-library cart constraint, ensuring users can only build and process orders from one specific library location per transaction to streamline hyper-local fulfillment.

To support offline logistics, the platform manages a delivery radius of 8km. The solution includes a comprehensive Admin application for each library to manage cataloging, track circulation workflows, and monitor availability. The system

dynamically enriches this catalog by automatically fetching author and publisher metadata via third-party integrations (such as the Open Library API). Finally, the platform handles fee management, issuing automated penalty notifications for overdue books and managing secure user authentication and password reset workflows via Nodemailer integration.

User Profiles & Actors

1. The Parent / Guardian (Primary Account Holder)

- **Description:** The adult responsible for creating and managing the household's account. They hold the secure credentials (verified via Nodemailer) and act as the financial and logistical manager for the family.
- **Access Level:** High. Accesses the PIN-gated "Parent Mode" within the mobile application.
- **Key Responsibilities & Capabilities:**
 - Creates and manages individual sub-profiles for their children.
 - Selects the primary local library branch for physical book delivery (within the 8km radius).
 - Manages the universal household Cart, ensuring compliance with the strict single-library checkout constraint.
 - Handles payments, views penalty notifications for overdue books, and manages delivery addresses.

2. The Child / Reader (Sub-Profile End User)

- **Description:** The primary consumer of the application's content, aged between 2 and 10 years old. They do not have login credentials, email addresses, or purchasing power.
- **Access Level:** Restricted. Operates entirely within the highly visual, secure "Kids Mode."
- **Key Responsibilities & Capabilities:**
 - Interacts with the AI-powered "virtual librarian" (RAG recommendation system) to discover new books based on their specific age rating and preferred genres.
 - Browses Author and Publisher pages (UC-25) to find notable books from their favorite creators.
 - **Constraint:** Cannot add books to the cart, check out, or modify account settings.

3. The Branch Librarian (Local Administrator)

- **Description:** The operational manager assigned to a specific physical micro-library or community book node.
- **Access Level:** Branch-Admin. Accesses the comprehensive web-based or tablet Admin application, restricted to their specific libraryId.
- **Key Responsibilities & Capabilities:**

- Manages local physical inventory, including cataloging new books (which auto-fetches Author/Publisher metadata via third-party APIs like Open Library).
- Processes the intake and outflow of physical books and tracks the 8km delivery logistics.
- Monitors overdue reservations and triggers automatic fee/penalty workflows for users.
- Ensures the physical stock aligns with the database so the AI recommendation engine only suggests available books to local children.

4. The System Administrator (Super Admin)

- **Description:** The platform owner or core developer managing the overarching cloud ecosystem.
- **Access Level:** Super-Admin. Has global read/write access across all collections in the MongoDB master database.
- **Key Responsibilities & Capabilities:**
 - Onboards new micro-library branches and assigns Branch Librarian credentials.
 - Monitors platform health, API rate limits (for external metadata fetching and Cloud Embeddings), and global transaction logs.
 - Manages platform-wide genre categories, age-rating definitions, and system updates.